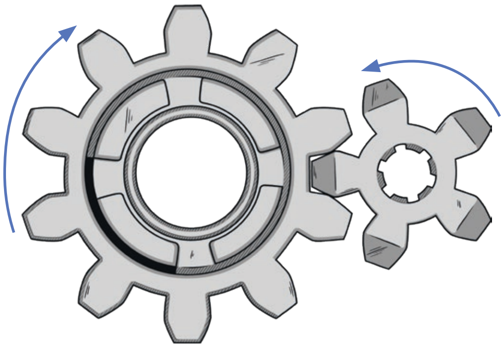
The Mechanic **Gear Ratios**

A gear ratio is a direct measure of the ratio of the rotational speeds of two or more interlocking gears. For example, the picture shows a 10 teeth gear and a 5 teeth gear.



1. Complete the table showing the gear ratios (the ratio of the driver gear to the driven gear). Do not simplify your answers.

Driver Gear	Driven Gear	Ratio
7	21	
10	2	
5		5:15
	5	25:5

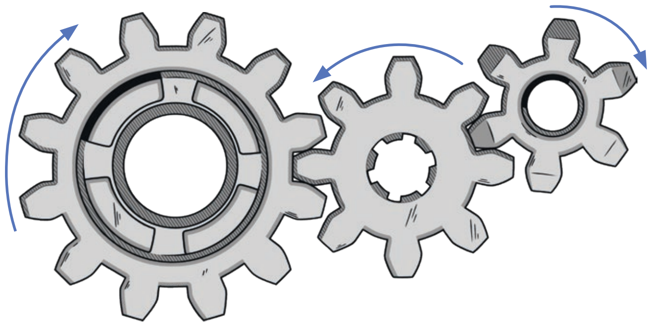
In the example showing a 10 teeth gear with a 5 teeth gear, if the 10 teeth gear is the driver then the ratio is 10:5. This simplifies to 2:1 and is known as a gearing up ratio. If the 5 teeth gear is the driver the ratio still simplifies to 2:1 but this is known as a gearing down ratio.

2. Complete the table to calculate the simplified gear ratio. In each case, state if it is a gearing up or gearing down ratio.

Driver Gear	Driven Gear	Ratio	Simplified Ratio	Gearing Up or Gearing Down?
3	4			
12	2			
5		5:25		
		8:	4:7	

A gear train consists of 2 or more gears. Only the first and last gears are important when calculating the gear ratio. The first is the driver gear and the last is the driven gear.

3. In a gear train, the driver gear has 12 teeth and the driven gear has 6. What would be the effect on the rotation of the gears if an 8 teeth gear was added to the middle of the gear train?

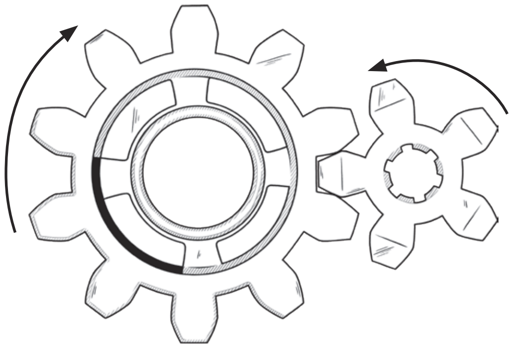


4. Prove that the gear ratio of the train shown in question three is the same as the gear ratio of a gear train where the driver gear has 12 teeth and the driven gear has 6.

   Hint: The intermediate gear ratios of a gear train will multiply together to equal the overall gear ratio.

The Mechanic **Gear Ratios Answers**

A gear ratio is a direct measure of the ratio of the rotational speeds of two or more interlocking gears. For example, the picture shows a 10 teeth gear and a 5 teeth gear.



1. Complete the table showing the gear ratios (the ratio of the driver gear to the driven gear). Do not simplify your answers.

Driver Gear	Driven Gear	Ratio
7	21	**7:21**
10	2	**10:2**
5	**15**	5:15
25	5	25:5

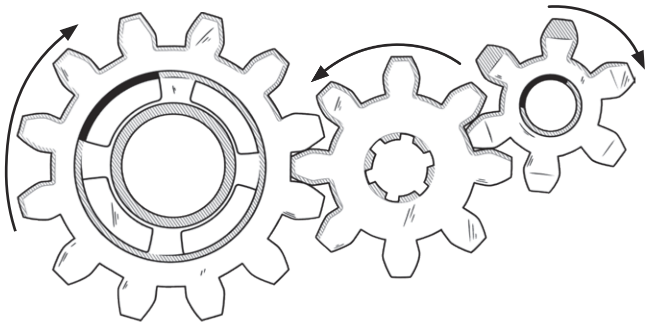
In the example showing a 10 teeth gear with a 5 teeth gear, if the 10 teeth gear is the driver then the ratio is 10:5. This simplifies to 2:1 and is known as a gearing up ratio. If the 5 teeth gear is the driver the ratio still simplifies to 2:1 but this is known as a gearing down ratio.

2. Complete the table to calculate the simplified gear ratio. In each case, state if it is a gearing up or gearing down ratio.

Driver Gear	Driven Gear	Ratio	Simplified Ratio	Gearing Up or Gearing Down?
3	4	**3:4**	**3:4**	**down**
12	2	**12:2**	**6:1**	**up**
5	**25**	5:25	**1:5**	**down**
8	**14**	8:**14**	4:7	**down**

A gear train consists of 2 or more gears. Only the first and last gears are important when calculating the gear ratio. The first is the driver gear and the last is the driven gear.

3. In a gear train, the driver gear has 12 teeth and the driven gear has 6. What would be the effect on the rotation of the gears if an 8 teeth gear was added to the middle of the gear train?



With the 12:6 gear train, the driver gear and driven gear will rotate in opposite directions. With a middle gear added, both the driver and driven gears will rotate in the same direction.

4. Prove that the gear ratio of the train shown in question three is the same as the gear ratio of a gear train where the driver gear has 12 teeth and the driven gear has 6.

   Hint: The intermediate gear ratios of a gear train will multiply together to equal the overall gear ratio.

   12:8:6

   12:8 simplifies to 3:2 and 8:6 simplifies to 4:3

   3 × 4 = 12, 2 × 3 = 6

   12:6 or 2:1